11th JEE RESTART 2026

1

Theoretical

Lecture - 01 | Physics

Waves

By- Neeraj Kr. Chaudhary

Topics to be covered

1. Introduction : Waves + Equation of Wave
2. Velocity of wave
3. Interference of waves
4. PYQs

Waves

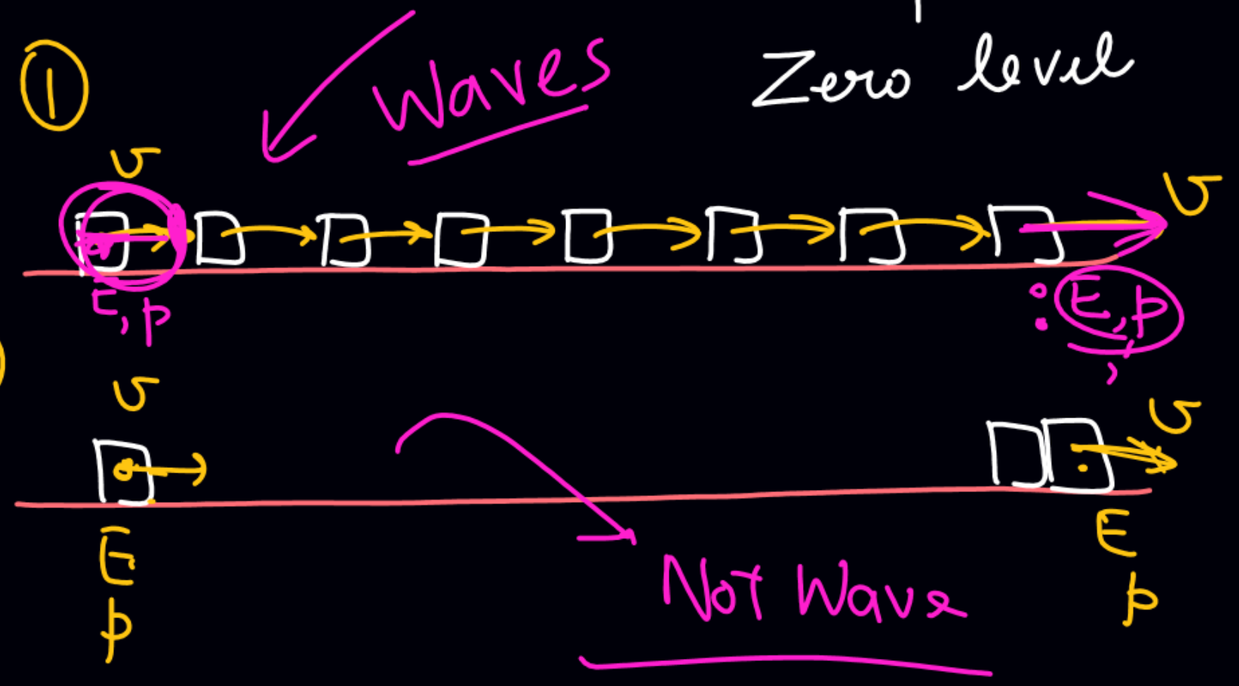
Energy + Momentum

1 → Transfer (without Transferring the mid) 2

Via Waves

Full Concepts

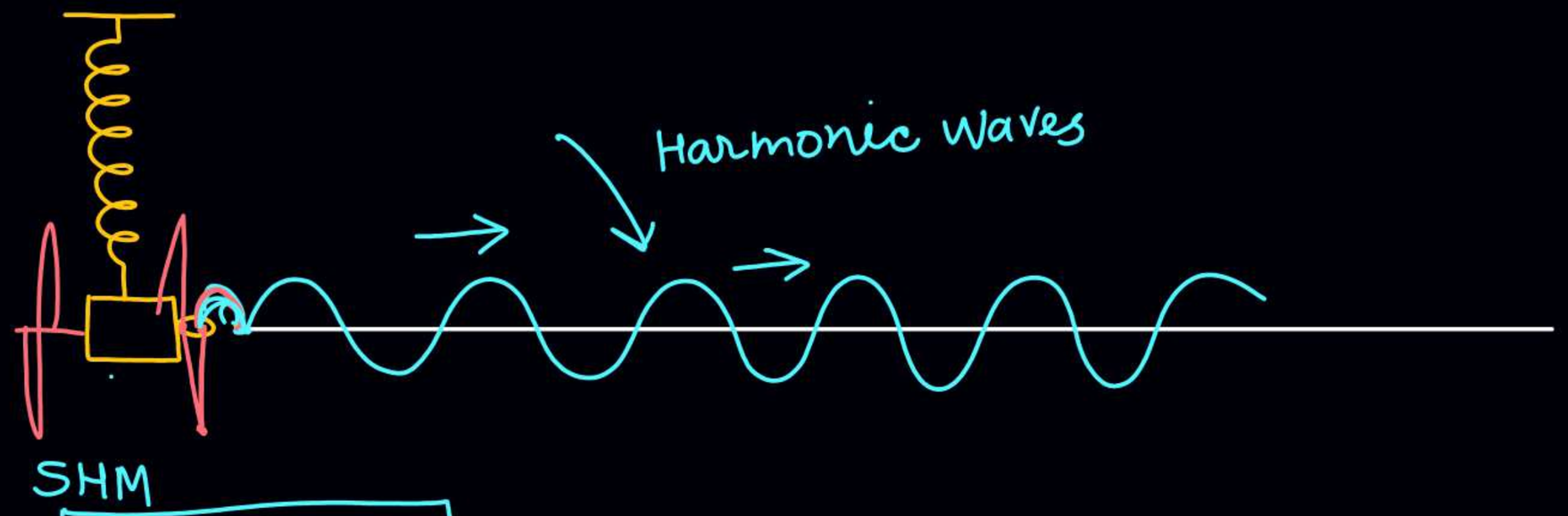
Zero level



* Wave Eq^n

JEE → Harmonic Waves

SHM → waves



$$\boxed{y = A \sin \omega t}$$

* Wave Eqn

$$\omega t_0 = \omega \cdot \frac{x}{v}$$
$$= \frac{\omega}{v} x$$
$$= Kx$$

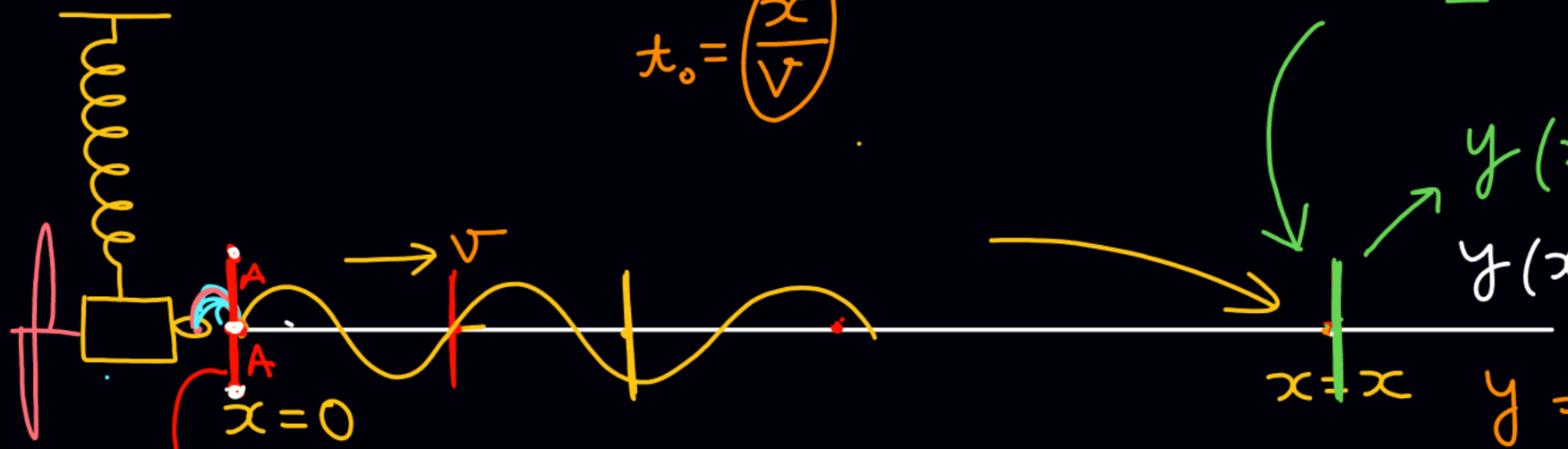
$(t - t_0)$

Re-Re Start

$V_{wave} = v$

$t_0 = \frac{x}{v}$

$t = t$



$y(x,t) = y(x=0, t-t_0)$

$y(x,t) = A \sin \omega (t - t_0)$

$y = A \sin(\omega t - \omega t_0)$

$y = A \sin\left(\omega t - \frac{\omega}{v} x\right)$

$$\boxed{y = A \sin(\omega t - Kx)}$$

SHM

$$\boxed{y = A \sin \omega t} \rightarrow y = A \sin \omega (t - t_0)$$

$$\boxed{y = A\sin(\omega t - Kx)}$$

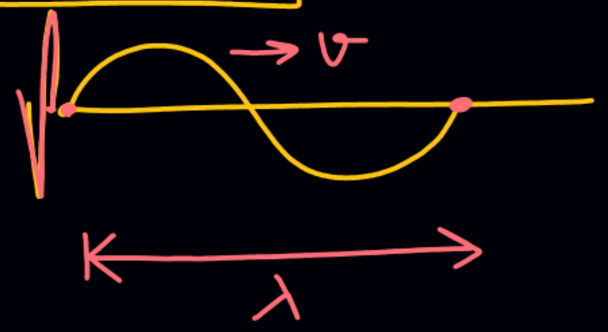
$$y = A\sin\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}\cdot x\right)$$

$$\boxed{y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)}$$

$$\boxed{K = \frac{\omega}{v}} = \frac{2\pi}{T}\cdot\frac{T}{\lambda} = \boxed{\frac{2\pi}{\lambda} = K}$$

$$\boxed{\omega = \frac{2\pi}{T}}$$

$$\boxed{T\cdot v = \lambda}$$



$$T = \frac{1}{f} \rightarrow \frac{2\pi}{\omega} = \frac{1}{f}$$

$$\boxed{\omega = 2\pi f}$$

Ang Freq.

Freq.

Q. Future

$t + dt$

$\boxed{a \propto -y}$

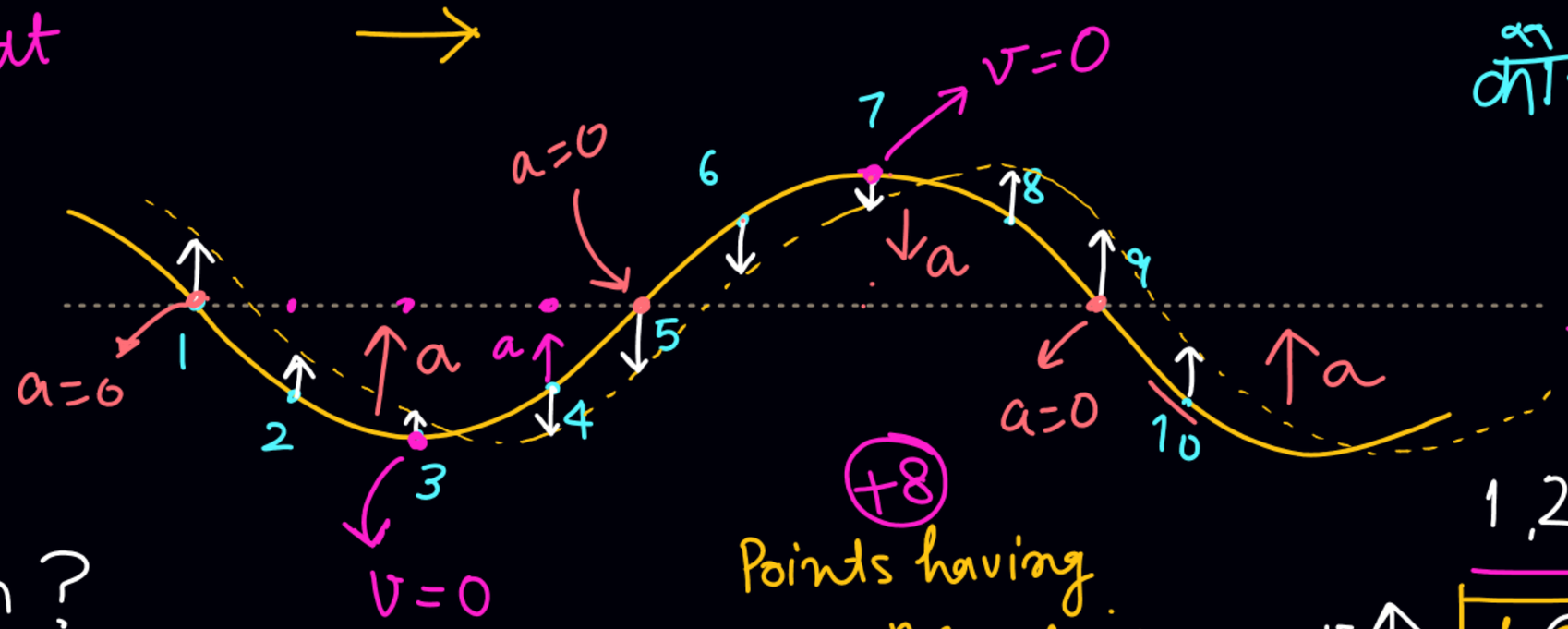
points acc^n & vel opposite = ?

(+4) marks.

कौन से Points ?

going up

$v \neq 0$



going down ?

Ans → 4567 (7 ×)

$\boxed{4, 5, 6}$ ↓ $\boxed{6, 7, 8}$ (a ↓)

(+8)

Points having acc^n & vel in same direction ?

$\boxed{6, 2, 10}$

Ans

1, 2, 3, 8, 9, 10

v ↑ $\boxed{1, 2, 8, 9, 10}$

a ↑ 2, 3, 4, 10

Q. Future

$t + dt$

$\boxed{a \propto -y}$

points acc^n & vel opposite = ? 4, 8

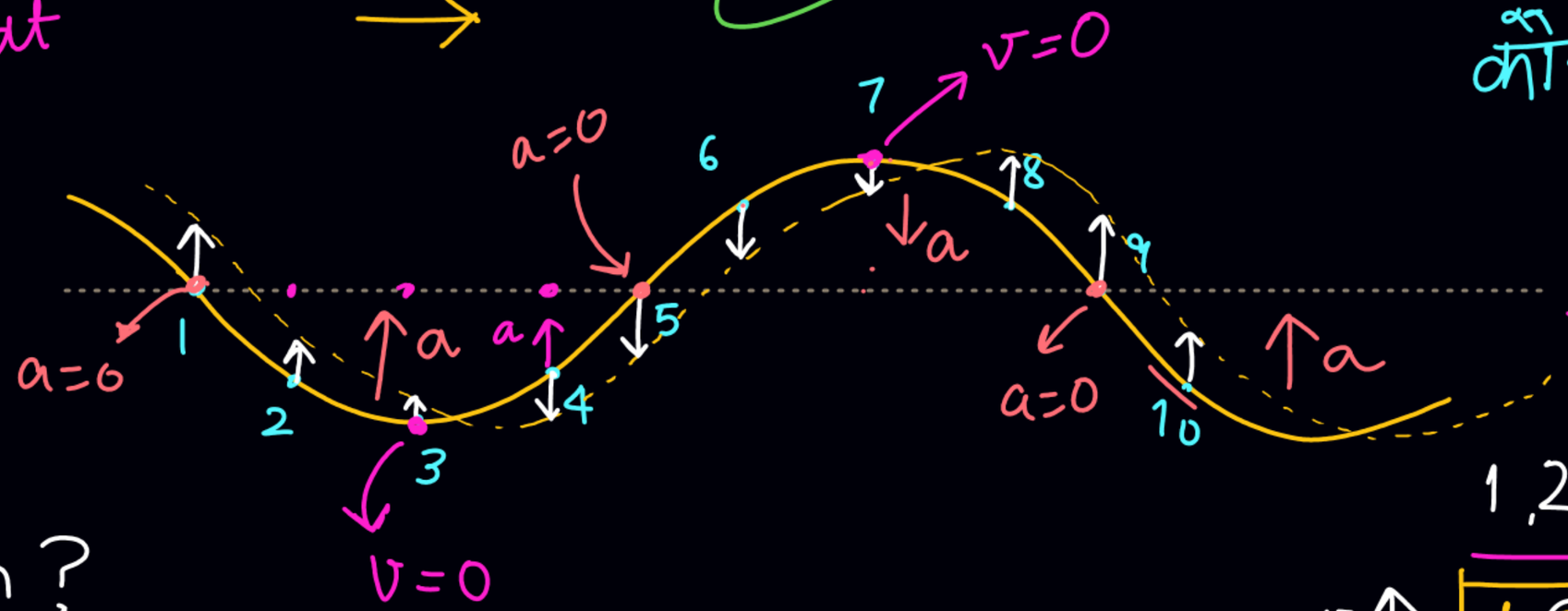
+4

+4 marks.

कौन से Points ?

going up

$v \neq 0$



going down ?

Ans → 4 5 6 7 ✗

$\boxed{4, 5, 6}$ ↓

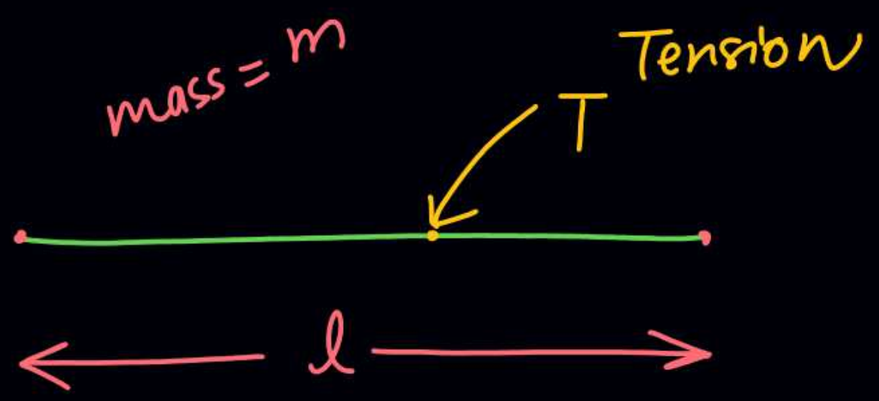
a↑ 2, 3, 4, 10

1, 2, 3, 8, 9, 10

v↑ $\boxed{1, 2, 8, 9, 10}$

a↓ $\boxed{6, 7, 8}$

Velocity of wave



$$\mu = \frac{\text{mass}}{\text{length}} = \frac{m}{l}$$

Velocity of wave

$$\boxed{V = \sqrt{\frac{T}{\mu}}}$$

$M_0 \gg m$

Q. $5gm \ll 10000\ gm$

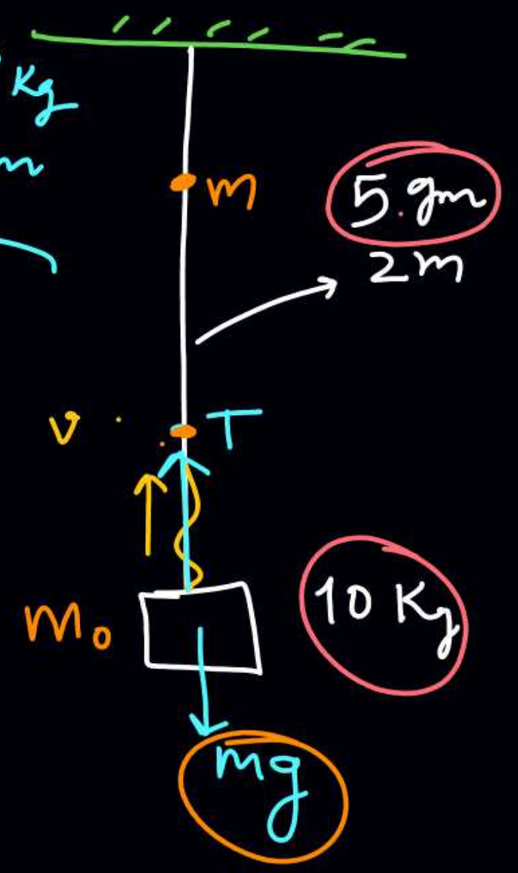
$$T = mg = 10 \times 10 = 100$$

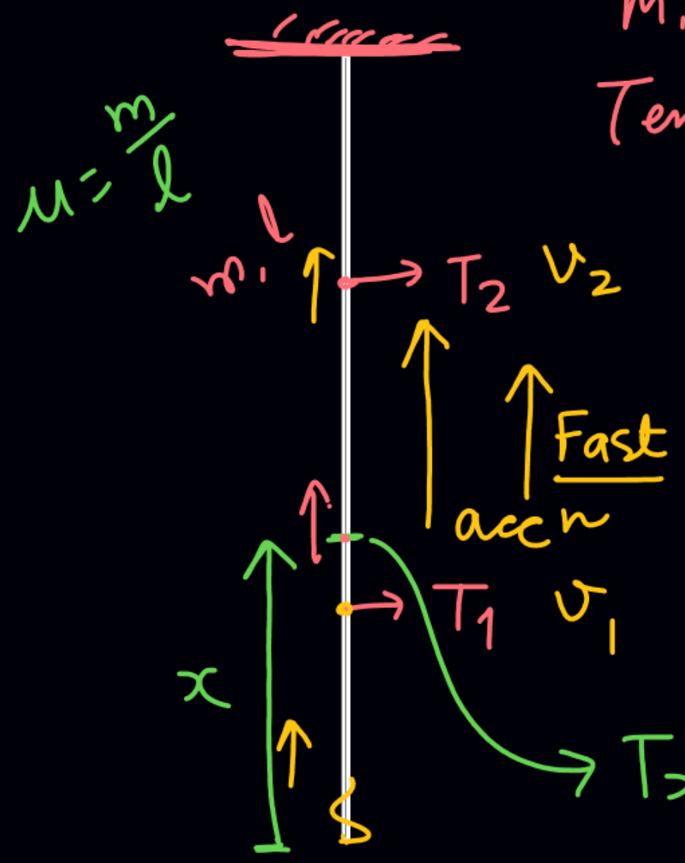
$$\mu = \frac{\text{mass}}{\text{length}} = \frac{5 \times 10^{-3}\ Kg}{2\ m}$$

$$V = \sqrt{\frac{100 \times 2}{5 \times 10^{-3}}}$$

$$= \sqrt{40 \times 10^{+3}}$$

$$V = 200\ m/s$$





M.L.

Tension due to its own mass.

$T_2 > T_1 \longrightarrow v_2 > v_1$

$v \propto \sqrt{T}$

acc^n = ?

$T_x = \frac{m}{l} x \cdot g$

$v_x = \sqrt{\frac{T_x}{\mu}} = \sqrt{\frac{\cancel{m} \cdot x g}{\cancel{l} \, \frac{\cancel{m}}{\cancel{l}}}}$

$$\boxed{V_x = \sqrt{xg}}$$

$V_x = \frac{dx}{dt}$

$a = \frac{dv}{dt}$

$V_x^2 = x g$

$2 \cdot \cancel{V_x} \cdot \left(\frac{dV_x}{dt}\right) = g \cdot \cancel{\frac{dx}{dt}}$

$$\boxed{a = \frac{g}{2}}$$

10:00 तक

कसम AIR की

No searching on internet

No cheating

100% ईमानदारी

Question

$x = A\sin(\omega t \pm \phi)$ $\times 4$

Displacement of a wave is expressed as $x(t) = 5\cos\left(628t + \frac{\pi}{2}\right)$ m. The wavelength of the wave when its velocity is 300 m/s is: **[JEE Main 2025]**

$= 5\sin(628t \pm \phi)$

1. 5 m
2. 3 m 37.5%
3. 0.5 m
4. 0.33 m

$\boxed{A = 5, \quad \omega = 628}$

$v = 300\,m/s$

$T = \frac{2\pi}{\omega} = \frac{6.28}{628} = \frac{1}{100}$

$\cos\theta = \sin\left(\theta \pm \frac{\pi}{2}\right)$

$\lambda = v * T$

$\lambda = 3\cancel{00} * \frac{1}{\cancel{100}}$

$\boxed{\lambda = 3}$

Ans: (B)

Question

Free

A plane progressive wave is given by $y = 2\cos 2\pi(330t - x)$m. The frequency of the wave is: **[JEE Main 2024]**

$y = A\cos 2\pi(330t - 1x)$.

$y = A\sin(\underbrace{2\pi * 330}\, t - \underbrace{2\pi}\, x \pm \frac{\pi}{2})$

1. 165 Hz
2. 330 Hz 52%
3. 660 Hz
4. 340 Hz

$f = \frac{1}{T} = \frac{\omega}{2\pi}$

$\omega = 2\pi * 330$

$f = \frac{\omega}{2\pi} = \frac{2\pi \times 330}{2\pi} = 330$ Hz

Ans: (B)

Question

The equation of a wave travelling on a string is $y = \sin[20\pi x + 10\pi t]$, where x and t are distance and time in SI units. The minimum distance between two points having the same oscillating speed is:

[JEE Main 2025]

1. 5.0 cm
2. 20 cm
3. 10 cm
4. 2.5 cm

$\frac{\lambda}{2}$

$\frac{10}{2} = 5cm$

$\frac{2\pi}{\lambda} = 20\pi$

$\lambda = \frac{1}{10} = 10cm$

$K = \frac{2\pi}{\lambda}$

$10\pi = \frac{2\pi}{\lambda}$

$\lambda = \frac{1}{5} = 20cm$

SHM

$V_P = \omega\sqrt{A^2 - x^2}$

ω, A Constt

x Same

Speed Same

Question

$kx + \omega t$

The equation of a wave travelling on a string is $y = \sin[20\pi x + 10\pi t]$, where x and t are distance and time in SI units. The minimum distance between two points having the same oscillating speed is:

[JEE Main 2025]

$K = 20\pi = \frac{2\pi}{\lambda} \rightarrow \lambda = 10$ cm

SHM

$$V_P = \omega\sqrt{A^2 - x^2}$$

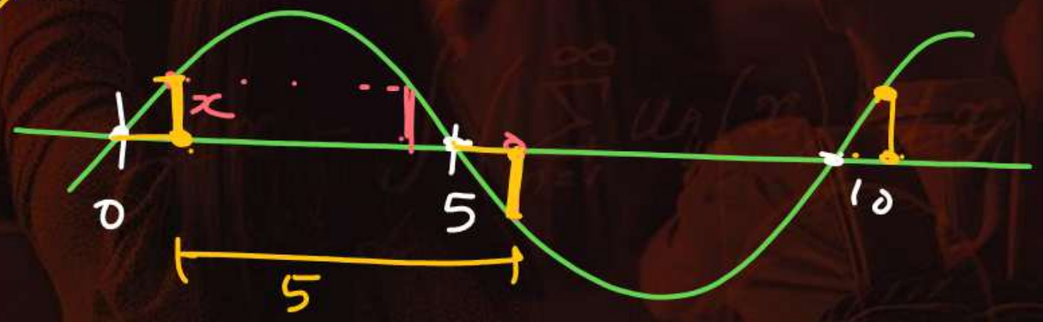
ω, A Constt

x - Same

Speed Same

1. 5.0 cm
2. 20 cm
3. 10 cm
4. 2.5 cm

Sure



5 Cm

Ans: (B)

Question

A steel wire with mass per unit length 7.0×10^{-3} kg/m is under tension of 70 N. The speed of transverse waves in the wire will be: **[JEE Main 2023]**

1. 200π m/s
2. 100 m/s 60%
3. 10 m/s
4. 50 m/s

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{70}{7 \times 10^{-3}}} = \sqrt{10^4} = 100$$

Ans: (B)

Question

समाप्त

In the wave equation $y = 0.5\sin\frac{2\pi}{\lambda}(400t - x)m$ the velocity of the wave will be:

[JEE Main 2022]

1. 200 m/s
2. $200\sqrt{2}$ m/s
3. 400 m/s
4. $400\sqrt{2}$ m/s

$y = .5 \sin \frac{2\pi}{\lambda}(400t - x)$

$v = 400$

$(x \pm vt)$

$(400t - 1x)$

vel

$t - t_0$

$\left(t - \frac{x}{v_0}\right)$

$\frac{1}{v_0}(v_0 t - x)$

Ans: (C)

Homework

IIT Dost 2008

Call help ??

IITian

Packa

Story

DPP & H.W.

String Wave

HCV - 14 - 32

2005 2006

IIT Delhi

2006 → Bansal Classes (Kota)

Thank You